Use case	Registration	
Release (id)	UC1, v0.1, 19/04/2026	
Context of Use	User authentication and account management.	
Scope	Let an anonymous user create a new account.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
Preconditions	The user must be logged out.	
Minimal Guarantees	No duplicated or unsecure account is created.	
Success Guarantees	The anonymous user successfully created an account and is logged in.	
Trigger	The anonymous user selects “register”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “register”.
	2	The system asks the user to insert username and email.
	3	The user inserts the required data.
	4	The system checks whether an account with at least one between username and email already exists.
	5	The system asks the user to insert the password twice.
	6	The user inserts the required data.
	7	The system checks whether the password satisfies the security requirements and if the two passwords coincide.

	8	The system asks the user's consent for personal data processing.
	9	The anonymous user accepts the personal data processing.
	10	The system saves the data in the account archive.
	11	The system notifies the user of successful registration with an email.
	12	The system switches to the authenticated user functionalities.
Extension	**Step**	**Other action**
	4a	Invalid username or email: UC1.1
	7a	Invalid password: UC1.2
Technology & data Variations	None	

Alternative flow	Registration: Invalid username or email	
Release (id)	UC1.1, v0.2, 25/04/2026	
Context of Use	User authentication and account management.	
Scope	The system alerts the anonymous user that she has inserted an invalid username or email.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
Preconditions	The user inserted an already existing username or email.	
Minimal Guarantees	No duplicated account is created.	
Success Guarantees	None	
Trigger	UC1	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid username or email.
	2	The sequence of events starts at step 2 of the main flow (UC1).
Technology & data Variations	None	

Alternative flow	Registration: Invalid password	
Release (id)	UC1.2, v0.2, 25/04/2026	
Context of Use	User authentication and account management.	
Scope	The system alerts the anonymous user that she has inserted an invalid password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	No unsecure account is created.	
Success Guarantees	None	
Trigger	UC1	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid password, specifying the problem.
	2	The sequence of events starts at step 5 of the main flow (UC1).
Technology & data Variations	None	

Use case	Account deletion	
Release (id)	UC2, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user delete her account.	
Level	Primary task	
Primary actor	Passenger or train company	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to delete her account
	Train company	Wants to delete its account
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The user successfully deleted her account and is logged out.	
Trigger	The user selects “delete account”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “delete account”.
	2	The system removes the data in the account archive.
	3	The system notifies the user of successful deletion with an email.
	4	The system switches to the unauthenticated user functionalities.
Extension	**Step**	**Other action**
	-	-
Technology & data Variations	None	

Use case	Account data view	
Release (id)	UC3, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user view her account's data.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager)	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to view her account's data
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The user successfully views her account's data.	
Trigger	The anonymous user selects "view account's data".	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality "view account's data".
	2	The system shows the data saved in the account archive, that is username and email.
Extension	**Step**	**Other action**
	-	-
Technology & data Variations	None	

Use case	Account data change	
Release (id)	UC4, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user change her account's data.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager)	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change her account's data
Preconditions	The user must be logged in.	
Minimal Guarantees	No duplicated or unsecure account is created with a modification.	
Success Guarantees	The authenticated user successfully changed her account's data.	
Trigger	The authenticated user selects "change data".	
Description	**Step**	**Action**
	1	The authenticated user selects the functionality "change data".
	2	The system asks the user which data to modify.
	3	If the user selects username
	3.1	The system asks the user to insert the new username.
	3.2	The user inserts the new username.
	3.3	The system checks whether an account with the same username already exists.
	4	If the user selects email
	4.1	The system asks the user to insert the new email.

	4.2	The user inserts the new email.
	4.3	The system checks whether an account with the same email already exists.
	5	If the user selects password
	5.1	The system asks the user to insert the password twice.
	5.2	The user inserts the required data.
	5.3	The system checks whether the password satisfies the security requirements and if the two passwords coincide.
	6	The system saves the data in the account archive.
	7	The system notifies the user of successful data change with an email.
Extension	**Step**	**Other action**
	3.3a	Invalid username or email: UC4.1
	4.3a	Invalid email: UC4.2
	5.3a	Invalid password: UC4.3
Technology & data Variations	None	

Alternative flow	Account data change: Invalid username	
Release (id)	UC4.1, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid username.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change username
Preconditions	The user inserted an already existing username.	
Minimal Guarantees	No duplicated account is created.	
Success Guarantees	None	
Trigger	UC4	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid username.
	2	The sequence of events starts at step 3.1 of the main flow (UC4).
Technology & data Variations	None	

Alternative flow	Account data change: Invalid email	
Release (id)	UC4.2, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid email.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change email
Preconditions	The user inserted an already existing email.	
Minimal Guarantees	No duplicated account is created.	
Success Guarantees	None	
Trigger	UC4	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid email.
	2	The sequence of events starts at step 4.1 of the main flow (UC4).
Technology & data Variations	None	

Alternative flow	Registration: Invalid password	
Release (id)	UC4.3, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid password.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change password
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	No unsecure account is created.	
Success Guarantees	None	
Trigger	UC4	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid password, specifying the problem.
	2	The sequence of events starts at step 5.1 of the main flow (UC4).
Technology & data Variations	None	

Use case	Reset password	
Release (id)	UC5, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	Let an anonymous user reset the password.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset the password
Preconditions	The user must be logged out.	
Minimal Guarantees	No unsecure account is created with a modification. No user can change the password of another user.	
Success Guarantees	The anonymous user successfully changed the password.	
Trigger	The anonymous user selects “forgot password”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “forgot password”.
	2	The system asks the user to insert the email associated to her account.
	3	The user inserts the email.
	4	The system checks if the email is associated to an existing account.
	5	The system creates a random identifier.
	6	The system sends the identifier to the email.
	7	The system asks the user to insert the identifier it sent.
	8	The user inserts the identifier.
	9	The system checks whether the identifier is correct.

	10	The system asks the user to insert the password twice.
	11	The user inserts the required data.
	12	The system checks whether the password satisfies the security requirements and if the two passwords coincide.
	13	The system saves the data in the account archive.
	14	The system notifies the user of successful password change with an email.
Extension	**Step**	**Other action**
	4a	Invalid email: UC5.1
	9a	Invalid identifier: UC5.2
	12a	Invalid password: UC5.3
Technology & data Variations	None	

Alternative flow	Reset password: Invalid email	
Release (id)	UC5.1, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid email.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset password
Preconditions	The user inserted an email that does not correspond to any existing account.	
Minimal Guarantees	None	
Success Guarantees	None	
Trigger	UC5	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid email.
	2	The sequence of events starts at step 2 of the main flow (UC5).
Technology & data Variations	None	

Alternative flow	Reset password: Invalid identifier	
Release (id)	UC5.2, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid identifier.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset password
Preconditions	The user inserted an identifier that does not correspond to the one created by the system.	
Minimal Guarantees	None	
Success Guarantees	None	
Trigger	UC5	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid identifier.
	2	The sequence of events starts at step 9 of the main flow (UC5).
Technology & data Variations	None	

Alternative flow	Reset: Invalid password	
Release (id)	UC5.3, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the anonymous user that she has inserted an invalid password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset password
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	No unsecure password is created.	
Success Guarantees	None	
Trigger	UC5	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid password, specifying the problem.
	2	The sequence of events starts at step 12 of the main flow (UC5).
Technology & data Variations	None	

Use case	Login	
Release (id)	UC6, v0.1, 20/04/2026	
Context of Use	User authentication.	
Scope	Let an anonymous user login.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to login
Preconditions	The user must be logged out.	
Minimal Guarantees	No user login in an account she has no authorization for.	
Success Guarantees	The anonymous user is logged in her account.	
Trigger	The anonymous user selects “login”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “login”.
	2	The system asks the user to insert username and password.
	3	The user inserts the required data.
	4	The system checks whether both the username and password correspond to an existing account.
	5	The system switches to the authenticated user functionalities.
Extension	**Step**	**Other action**
	4a	Invalid username or password: UC5.1
Technology & data Variations	None	

Alternative flow	Login: Invalid username or password	
Release (id)	UC6.1, v0.2, 25/04/2026	
Context of Use	User authentication.	
Scope	The system alerts the anonymous user that she has inserted an invalid username or password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to login
Preconditions	The user inserted a pair of username and password that does not correspond to any existing account	
Minimal Guarantees	No user login in an account she has no authorization for.	
Success Guarantees	None	
Trigger	UC5	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid username or password, specifying the problem.
	2	The sequence of events starts at step 2 of the main flow (UC5).
Technology & data Variations	None	

Use case	Logout	
Release (id)	UC7, v0.1, 20/04/2026	
Context of Use	User authentication.	
Scope	Let an authenticated user logout.	
Level	Primary task	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to logout
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The authenticated user is logged out.	
Trigger	The anonymous user selects “logout”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “logout”.
	5	The system switches to the anonymous user functionalities.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket display	
Release (id)	UC8, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Let a user view the details of a ticket.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager) or anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to view tickets' details
	Anonymous user	Wants to view tickets' details
Preconditions	The user must be in an interface where is displayed a ticket. ???fix?	
Minimal Guarantees	None	
Success Guarantees	The user successfully view a ticket's details.	
Trigger	The user selects "view tickets' details".	
Description	**Step**	**Action**
	1	The authenticated user selects the functionality "view tickets' details".
	2	The system displays the ticket's details.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket purchase	
Release (id)	UC9, v0.2, 25/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to purchase a ticket.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to buy a ticket
	Payment gateway	Manages transaction
Preconditions	The passenger must be logged in. The passenger must be in an interface displaying the ticket's details.	
Minimal Guarantees	The passenger who could not pay did not purchase any ticket.	
Success Guarantees	The passenger successfully purchased a ticket.	
Trigger	The passenger selects "Buy ticket".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Buy ticket".
	2	The system asks the user confirmation for the purchase.
	3	The user confirms the purchase.
	4	The system sends a transaction request to the payment gateway.
	5	The payment gateway manages the transaction.
	6	The payment gateway sends the transaction's status to the system.
	7	If the transaction is not successful

	7.1	The system alerts the passenger about the failed transaction.
	8	Otherwise
	8.1	The system alerts the passenger about the completed transaction.
	8.2	The system notifies the passenger of successful purchase with an email.
	8.3	The system saves the ticket in the account archive.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket cancellation	
Release (id)	UC10, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to cancel a purchased ticket.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to buy a ticket
	Payment gateway	Manages transaction
Preconditions	The passenger must be logged in. The passenger must be in an interface displaying the ticket's details. The passenger must have already purchased the ticket.	
Minimal Guarantees	The passenger does not cancel any ticket she has not purchased.	
Success Guarantees	The passenger successfully cancelled a purchased ticket.	
Trigger	The passenger selects "Cancel ticket".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Cancel ticket".
	2	The system asks the user confirmation for the cancellation.
	3	The user confirms the cancellation.
	4	The system sends a transaction request to the payment gateway.
	5	The payment gateway manages the transaction.
	6	The payment gateway sends the transaction's status to the system.

	7	If the transaction is not successful
	7.1	The system alerts the passenger about the failed transaction.
	8	Otherwise
	8.1	The system alerts the passenger about the completed transaction.
	8.2	The system notifies the passenger of successful purchase with an email.
	8.3	The system saves the ticket's cancellation information in the account archive.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket history	
Release (id)	UC11, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to view her tickets' history, both purchased and cancelled.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to view tickets' history
Preconditions	The passenger must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The passenger successfully view tickets' history.	
Trigger	The passenger selects "Tickets' history".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Tickets' history".
	2	The system shows the list of tickets purchased and cancelled by that passenger.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Find Path	
Release (id)	UC11, v0.1, 25/04/2026	
Context of Use	Passenger utility	
Scope	Let an anonymous user or passenger find a path between two stations	
Level	Primary task	
Primary actor	Anonymous user OR Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user OR Passenger	Find a path
Preconditions	The user must be either logged out or logged in as a Passenger	
Minimal Guarantees	No impossible path is proposed	
Success Guarantees	The user has a suitable path for other operations, that can be directly selected by the user for it.	
Trigger	The user selects “find a path”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “find a path”.
	2	The system asks the user to insert a Departing and Arriving station
	3	The user inserts all the requested data.
	4	The system ask the user an Departure or Arrival time
	5	The user provides any of the requested data, possibly confirming zero
	6	The system checks whether at least one path along the mesh between the two stations exists

	7	The system computes whether one or the intersection of multiple service lines (referred to as candidate paths) connects the stations
	8	The system filters out all candidate paths that do not respect Departure and Arrival times, if provided
	9	The system presents the user an option to change path-sorting algorithm from a default option
	10	The system uses the path-sorting algorithm to provide the top-ranking paths computed at step 8
	11	The user selectes one path
	12	The system discards all non-selected paths, and revokes access to the path-sorting algorithm switch
	13	The system presents to the user all path-specific actions regarding that path
Extension	**Step**	**Other action**
	3a	If the user does not insert sufficient data (leaves arrival or departure station blank), display “no arrival/departure station” error and return to step 2
	5a	If the user inserts zero data, default to “Departure” as current time, proceed normally
	6a	If no path is found, produce “Stations disconnected” error and return to step 2
	7a	If no candidate paths are found, produce “No path available” error and return to step 2
	8a	If no candidate paths remain after the filter, display “Too strict Arrival/Departure” error and return to step 4
	9-10a	If the user changes sorting algorithm, the newly selected one is used to repeat from step 8

	9-10b	If two or more paths score equally in selected sorting algorithm, use all other sorting algorithms in a pre-determined order as tie-breakers, and failing that, assign order among them randomly. Proceed as normal
Technology & data Variations	If user is not a logged-in Passenger, ticket actions for the path are displayed but marked as unavailable	